

WASTEWISE

Development of a Web and Mobile-Based Waste Collection Monitoring and Management System for Local Government Units

System Installation and Deployment Guide

Platform

Web and Mobile-Based Application

Technologies Used

- Node.js
 - Express.js
 - React.js
 - React Native / Expo
 - MongoDB Atlas
 - Google Maps API
 - Gmail SMTP
 - WebSocket Technology
-

WasteWise System Installation and Deployment Guide

1. System Overview

WasteWise is composed of three separate applications:

- Backend (Node.js / Express.js)
- Frontend (Web Application)
- Mobile Application (React Native / Expo)

The system uses:

- MongoDB Atlas for database storage
 - Google Maps API for map functionality
 - Gmail SMTP for OTP email notifications
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2. GitHub Repositories

Backend Repository

```
https://github.com/kapetstone333/WasteWiseBackend.git
```

Frontend Repository

```
https://github.com/kapetstone333/WasteWiseFrontend.git
```

Mobile Repository

```
https://github.com/kapetstone333/WasteWiseMobile.git
```

3. Database Configuration

MongoDB Compass Installation

Download MongoDB Compass:

```
https://www.mongodb.com/try/download/compass
```

Connect to the Database

1. Open MongoDB Compass.
2. Click Add New Connection (+).
3. Enter the following URI:

```
mongodb+srv://kapetstone_db_user:test@cluster0.kz4wjgy.mongodb.net/
```

1. Click Save & Connect.
 2. Verify that the database named `waste_wise` is accessible.
-

MongoDB Connection Used by the System

Create a `.env` file in the root directory of the Backend project.

File Location:

```
WasteWiseBackend/.env
```

Add the following:

```
MONGO_URI=mongodb+srv://kapetstone_db_user:test@cluster0.kz4wjgy.mongodb.net/waste_wise?retryWrites=true&w=majority&appName=Cluster0
```

```
PORT=5000
```

```
EMAIL_USER=kapetstone@gmail.com
```

```
EMAIL_PASS=zeozlrodklfwbslz
```

MongoDB Connection Source Code

File:

```
WasteWiseBackend/src/config/connect.js
```

Connection Code:

```
mongoose.connect(process.env.MONGO_URI)
```

If Migrating to Another Database Server

Only update:

```
MONGO_URI=
```

inside:

```
WasteWiseBackend/.env
```

4. Email OTP Configuration

The WasteWise system sends OTP emails using Gmail SMTP.

Configuration File

```
WasteWiseBackend/.env
```

Variables:

```
EMAIL_USER=kapetstone@gmail.com  
EMAIL_PASS=zeozlrodklfwbslz
```

OTP Mailer Source Code

File:

```
WasteWiseBackend/src/mailer/otp_mailer.js
```

Uses:

```
process.env.EMAIL_USER  
process.env.EMAIL_PASS
```

If Changing the Email Account

Update:

```
EMAIL_USER=  
EMAIL_PASS=
```

with the new Gmail account credentials and App Password.

5. Backend Installation

Clone the Repository

```
git clone https://github.com/kapetstone333/WasteWiseBackend.git
```

Install Dependencies

```
npm install
```

Configure Environment Variables

Create:

```
.env
```

inside:

```
WasteWiseBackend
```

Add:

```
MONGO_URI=mongodb+srv://kapetstone_db_user:test@cluster0.kz4wjgy.mongodb.net/waste_wise?retryWrites=true&w=majority&appName=Cluster0
```

```
PORT=5000
```

```
EMAIL_USER=kapetstone@gmail.com
```

```
EMAIL_PASS=zeozlrodklfwbslz
```

Run Backend

```
npm run dev
```

Backend Entry Point:

```
WasteWiseBackend/src/server.js
```

Expected URL:

```
http://localhost:5000
```

6. Frontend Installation

Clone the Repository

```
git clone https://github.com/kapetstone333/WasteWiseFrontend.git
```

Install Dependencies

```
npm install
```

Configure Backend URL

File:

```
WasteWiseFrontend/src/services/axios_instance.js
```

Current Configuration:

```
baseUrl: 'http://localhost:5000'
```

If Backend is Deployed on Another Server

Example:

```
baseUrl: 'https://your-backend-domain.com'
```

Replace the URL with the actual backend server URL.

Configure Google Maps API

File:

```
WasteWiseFrontend/index.html
```

Locate:

```
<script  
src="https://maps.googleapis.com/maps/api/js?  
key=YOUR_API_KEY&libraries=marker&loading=async"  
async  
defer>  
</script>
```

Replace:

```
YOUR_API_KEY
```

with the active Google Maps API key.

Run Frontend

```
npm run dev
```

7. Mobile Application Installation

Clone the Repository

```
git clone https://github.com/kapetstone333/WasteWiseMobile.git
```

Install Dependencies

```
npm install
```

Configure Backend URL

File:

```
WasteWiseMobile/services/axios_instance.js
```

Current Configuration:

```
baseUrl: 'http://localhost:5000'
```

If Backend is Deployed on Another Server

Example:

```
baseUrl: 'https://your-backend-domain.com'
```

Replace the URL with the actual backend server URL.

Configure Google Maps API

File:

```
WasteWiseMobile/app.json
```

Locate:

```
"apiKey": "YOUR_API_KEY"
```

Replace:

```
YOUR_API_KEY
```

with the active Google Maps API key.

Run Mobile Application

```
npm run start
```

8. Files to Update When Migrating to Another Server

Configuration	File Location
MongoDB Connection	WasteWiseBackend/.env
Email OTP Configuration	WasteWiseBackend/.env
Frontend Backend URL	WasteWiseFrontend/src/services/axios_instance.js
Mobile Backend URL	WasteWiseMobile/services/axios_instance.js
Frontend Google Maps API	WasteWiseFrontend/index.html
Mobile Google Maps API	WasteWiseMobile/app.json

9. Deployment Checklist

- ✓ MongoDB Atlas is accessible
- ✓ MongoDB URI is configured
- ✓ .env file exists
- ✓ EMAIL_USER is configured
- ✓ EMAIL_PASS is configured
- ✓ Backend server is running
- ✓ Frontend baseURL is correct
- ✓ Mobile baseURL is correct
- ✓ Google Maps API key is configured
- ✓ npm dependencies are installed
- ✓ No startup errors are present

10. Troubleshooting Guide

MongoDB Connection Error

Possible Causes:

- Invalid MONGO_URI
- Database server unavailable
- MongoDB Atlas network restrictions

Solution:

1. Verify MONGO_URI.
2. Verify internet connection.
3. Verify Atlas Network Access settings.
4. Verify MongoDB Atlas cluster is running.

Backend Not Starting

Possible Causes:

- Missing .env file
- Invalid environment variables
- Missing npm packages

Solution:

1. Verify .env file exists.
2. Verify MONGO_URI, EMAIL_USER, and EMAIL_PASS.

3. Run `npm install`.
4. Check terminal logs for errors.

Frontend Cannot Connect to Backend

Possible Causes:

- Backend not running
- Incorrect baseURL

Solution:

1. Verify backend server is running.
2. Verify `src/services/axios_instance.js`.
3. Verify the backend URL is accessible.

Mobile Cannot Connect to Backend

Possible Causes:

- Incorrect baseURL
- Backend unavailable

Solution:

1. Verify `services/axios_instance.js`.
2. Verify backend URL is accessible.
3. Verify backend service is online.

Google Maps Not Loading

Possible Causes:

- Invalid API key
- Google Maps API not enabled

Solution:

1. Verify Google Maps API key.
2. Enable Maps JavaScript API.
3. Verify Google Cloud billing configuration.

OTP Email Not Sending

Possible Causes:

- Invalid EMAIL_USER
- Invalid EMAIL_PASS
- Expired Gmail App Password

Solution:

1. Verify EMAIL_USER.
 2. Verify EMAIL_PASS.
 3. Generate a new Gmail App Password if necessary.
 4. Restart the backend service.

11. Turnover Notes

For future maintenance or deployment, the following configurations are the most important:

- MongoDB Connection → `WasteWiseBackend/.env`
- Email OTP Configuration → `WasteWiseBackend/.env`
- Frontend API Connection → `WasteWiseFrontend/src/services/axios_instance.js`
- Mobile API Connection → `WasteWiseMobile/services/axios_instance.js`
- Frontend Google Maps API → `WasteWiseFrontend/index.html`
- Mobile Google Maps API → `WasteWiseMobile/app.json`

If the system is moved to a new server, update these files first before running the application.

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